JEFREY BONYAD, ECE, ECT

Software & Systems Integration Engineer | MQTT | REST APIs

Embedded & Backend Development

San Nicolas, Sta Ana, Pampanga, Philippines | 09501057592 / 09057376078

jefreytiglaobonyad@gmail.com

Portfolio: jefrey-portfolio.vercel.app

GitHub: github.com/bbystps

LinkedIn: linkedin.com/in/jefrey-bonyad-592032250

PROFESSIONAL SUMMARY

Developer with experience connecting embedded devices, web applications, backend services, and MySQL databases. Built a Python MQTT backend on AWS, Laravel REST APIs, and real-time monitoring dashboards. Combines API consumption, Postman testing, server-side validation, and guided IoT learning workflows with a background in semiconductor production automation to support integration development and troubleshooting.

RELEVANT TECHNICAL SKILLS

Integration: MQTT, REST API development and API consumption, WebSocket, embedded-to-web integration, RS485 sensor communication, backend integration

Programming and data: Python, PHP, Laravel, Node.js, JavaScript, C/C++, MySQL, SQL, database design, CRUD operations, authentication, server-side validation

Testing and operations: Postman API testing, transaction logging, sensor and device monitoring, troubleshooting, Git, GitHub, AWS EC2, Linux, PuTTY, WinSCP

Embedded and interfaces: ESP32, Raspberry Pi, Arduino, FreeRTOS, sensor integration, React.js, HTML, CSS

Technical documentation: guided IoT exercises covering MQTT communication, sensor monitoring, device control, and firmware updates

PROFESSIONAL EXPERIENCE

Web, IoT & Embedded Systems Developer

Innovation Central PH | Full-time | January 2023 - Present | Remote, Philippines

- Develop backend applications and REST APIs using Laravel, PHP, Node.js, and MySQL; connect frontend applications to APIs and database-driven services.
- Implement server-side validation, authentication, database relationships, file uploads, and CRUD workflows; use Postman for API development and testing.
- Integrate web applications with MQTT, WebSocket, Raspberry Pi, ESP32, sensors, and controllers for real-time data acquisition, monitoring, and device control.
- Build monitoring dashboards and kiosk workflows; deploy and maintain applications on Linux and cloud servers using Git and GitHub for source control.

Information Technology Specialist

RNM Dynamics Philippines | Full-time | December 2019 - June 2023

- Developed and maintained company web applications, queueing, service-management, and file-storage systems using PHP, MySQL, JavaScript, Bootstrap, jQuery, and AJAX.
- Integrated web applications with hardware, sensors, controllers, and real-time communication for semiconductor production automation and monitoring.
- Collaborated across teams on production and automation solutions; maintained local network connectivity, firewalls, email servers, and cloud servers.

SELECTED TECHNICAL PROJECTS

Project SEDI Dashboard / Dam Monitoring System

Python | MQTT | AWS | IoT | 4G | Web dashboard

- Built a Python MQTT backend, deployed it on an AWS server, and developed a multi-site dashboard for geophone, turbidity, and water-level data.
- Added alerts, historical charts, and system-health indicators to support monitoring and visibility into connected sensor systems.

IoT Interactive Platform

November 2025 - January 2026 | HTML | CSS | JavaScript | MQTT | WebSockets | ESP32

- Developed a web-based learning platform connecting real hardware and live dashboards, with dedicated tracks for WiFi, GSM, LoRa, and BLE.
- Created guided exercises covering MQTT communication, sensor monitoring, dashboard control, RFID access, SMS alerts, and OTA firmware updates, providing a structured environment for hands-on IoT learning.

VendoLabsPH Product Management and Customer Website

React.js | Tailwind CSS | Laravel | PHP | MySQL | REST API | Postman | Git

Website: vendolabsph.com

- Developed a customer website and administrative dashboard backed by Laravel REST API endpoints for creating, retrieving, updating, and deleting product records.
- Connected React API requests to a MySQL product model covering SKU, slug, pricing, inventory, status, and images; implemented image upload, validation, filtering, and search. Tested backend APIs with Postman.

CardiHoop ECG Monitoring System

ESP32 | AD8232 | MQTT | Python | Edge Impulse | Database | Web dashboard

- Developed an ECG acquisition and monitoring workflow connecting ESP32 sensor data through MQTT to Python processing, Edge Impulse inference, database storage, and web-based scan records.
- Integrated heart-rate tracking and rhythm classification to support review of possible abnormal rhythms in athlete/patient scan records.

EDUCATION AND CREDENTIALS

Bachelor of Science in Electronics and Communication Engineering

Don Honorio Ventura State University | 2012 - 2017

Electronics Engineer Licensure Examination Passer | October 2017

Electronics Technician Licensure Examination Passer | October 2017

AWSome Day Online Conference | Certificate of Attendance | July 11, 2024